Arista Access Node WFD-ACC-31 QoS Congestion Event Alarm SOP (ALARM_7343)

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_7343

Description

This Standard Operating Procedure (SOP) addresses a Warning severity QoS Congestion Event alarm, specifically ALARM_7343, on the Arista Access node WFD-ACC-31 located at the Whitefield POP. The alarm is triggered due to Microbursts, interface tail drops, or QoS queue starvation. On Arista equipment, QoS is implemented using the Modular QoS CLI (MQC), which classifies and polices traffic based on predefined policies. The root cause of this alarm may be attributed to misconfigured QoS policies, inadequate policing mechanisms, or congested network interfaces. The Access node's role in this scenario involves managing and controlling the flow of traffic between the Whitefield POP and the rest of the network. In this context, the Access node's QoS configuration must be carefully managed to prevent congestion events and ensure optimal network performance. This SOP provides a step-by-step approach to diagnose and resolve the root cause of the ALARM_7343, ensuring that the Access node's QoS configuration is optimized and the network remains stable.

Pre-requisite Checks

- Verify that all redundant links are functioning correctly and that there are no known issues with the network infrastructure.

- Check the temperature and humidity levels at the Whitefield POP to ensure they are within the recommended operating range.

- Confirm that the Arista Access node WFD-ACC-31 is powered on and all interfaces are online.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate policing mechanisms	Medium
Congested network interfaces	Low

Diagnostic Commands & Expected Outputs

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Command: show qos policy, Expected Output: Display the QoS policy configuration for the Access node., Description: This command displays the current QoS policy configuration, including all defined policies, classes, and interfaces. Review the output to identify any misconfigured or missing policies that may be contributing to the congestion event.

Command: show interface, Expected Output: Display the interface status and statistics for the Access node., Description: This command
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displays the status and statistics for all interfaces on the Access node, including errors, drops, and utilization. Review the output to identify any interfaces experiencing congestion or errors.

Command: show mpls qos, Expected Output: Display the MPLS QoS configuration for the Access node., Description: This command displays the MPLS QoS configuration, including the assigned labels and forwarding classes. Review the output to identify any misconfigured or missing labels or forwarding classes that may be contributing to the congestion event.
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Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: copy running-config startup-config && reboot, Description: Save the current QoS policy configuration to the startup configuration and reboot the Access node to ensure the changes take effect. After the reboot, verify that the QoS policy configuration is correct and functioning as expected.

Step 2: Step: Adjust policing mechanisms, Command: policer cir bc pir be , Description: Adjust the policing mechanisms to ensure that traffic is properly policed and congested interfaces are cleared. Verify that the policing mechanisms are correctly configured and functioning as expected.

Step 3: Step: Clear congested interfaces, Command: clear counters interface , Description: Clear the counters for the congested interfaces to ensure accurate monitoring and prevent further congestion events. Verify that the counters are cleared and the interfaces are functioning correctly.

Post-Check Verification

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Command: show qos policy, Expected Output: Display the QoS policy configuration for the Access node., Description: Verify that the QoS policy configuration is correct and functioning as expected.

Command: show interface, Expected Output: Display the interface status and statistics for the Access node., Description: Verify that the interface status and statistics are correct and functioning as expected.

Command: show mpls qos, Expected Output: Display the MPLS QoS configuration for the Access node., Description: Verify that the MPLS QoS configuration is correct and functioning as expected.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy startup-config running-config, Description: Restore the original QoS policy configuration by copying the startup configuration back to the running configuration. This will undo any changes made during the resolution process and return the Access node to its pre-resolution state.

- Step: Reboot the Access node, Command: reboot, Description: Reboot the Access node to ensure that the changes are reflected in the configuration. This will also clear any temporary changes made during the resolution process.